

Notes: Grossman & Rossi-Hansberg (AER, 2008)

Trading Tasks: A Simple Theory of Offshoring

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1 Big picture

- **Question.** How does a decline in the cost of offshoring affect domestic wages and welfare when firms trade tasks rather than complete goods?
- **Core departure from standard trade theory.** Traditional trade models focus on goods crossing borders. Grossman and Rossi-Hansberg model production as a continuum of tasks, some of which can be moved offshore while final goods production remains organized by home firms.
- **Main conceptual result.** Offshoring tasks creates three wage effects: a **productivity effect**, a **relative-price effect**, and a **labor-supply effect**.
- **Productivity effect.** When firms can perform some tasks abroad more cheaply, the factor whose tasks are offshored becomes effectively more productive at home. This can raise the domestic wage of that factor.
- **Contrast with goods trade.** Goods trade usually creates distributional conflict through Stolper–Samuelson forces. Task trade can generate shared gains because offshoring reduces production costs in a way similar to factor-augmenting technical change.
- **Contrast with immigration.** Immigration expands the domestic supply of a factor and therefore creates a labor-supply effect. Offshoring is different because it also creates inframarginal cost savings on tasks already offshored, producing the productivity effect.
- **Small open economy result.** In a two-good, two-factor small Heckscher–Ohlin economy with low-skill task offshoring, a fall in offshoring costs is isomorphic to low-skill-labor-augmenting technological progress, and the real wage of low-skilled labor rises.
- **Large economy result.** In a large economy, the productivity effect remains but is accompanied by relative-price changes. The relative price of the labor-intensive good can fall, which may hurt low-skilled labor through the Stolper–Samuelson channel.
- **Specialization result.** If the home country produces only one good or has more factors than goods, offshoring also frees workers from previously domestic tasks. The resulting labor-supply effect can reduce the wage of the offshored factor.
- **Skill-task extension.** If high-skill tasks can also be offshored, falling high-skill offshoring costs generate a productivity effect for skilled workers as well.
- **Takeaway.** Offshoring is not simply equivalent to importing goods or adding foreign workers. It changes the productivity of domestic factors by changing which tasks are performed at home and abroad.

2 Incremental contribution of the paper

2.1 Contribution relative to standard trade-in-goods theory

- Standard trade models focus on goods, sectors, and factor content. In these models, lower trade costs for goods typically change relative goods prices and redistribute income across factors.
- Grossman and Rossi-Hansberg instead model trade in tasks. A final good is produced by many tasks, and each factor supplies a continuum of tasks.
- This shift changes the logic of distribution. When the cost of moving some tasks abroad falls, the domestic factor associated with those tasks can become more productive, rather than simply more exposed to import competition.

Interpretation: The paper’s first contribution is to show that offshoring can be factor-augmenting technological progress in disguise. This is not visible in a model where countries only trade complete goods.

2.2 Contribution relative to fragmentation and intermediate-input models

- Earlier fragmentation and intermediate-input models recognized that production can be split across locations.
- Many of those models treat fragments as discrete and industry-specific, or assume uniform trade costs across fragments.
- This paper introduces a continuum of tasks ordered by heterogeneous offshoring costs. Firms choose a marginal task I , with tasks below I performed abroad and tasks above I performed at home.
- The continuum representation makes the extent of offshoring a continuous variable, allowing comparative statics and decomposition.

Interpretation: The model turns offshoring from a discrete sectoral event into a continuous reorganization of production, which makes it possible to identify general effects rather than case-by-case outcomes.

2.3 Contribution relative to offshoring-as-labor-supply views

- A common intuition is that offshoring low-skill tasks is equivalent to adding foreign low-skilled workers to the domestic labor market.
- Grossman and Rossi-Hansberg show why this is incomplete. Offshoring does release domestic workers from tasks that move abroad, but it also creates cost savings on tasks that were already offshored.
- These inframarginal cost savings raise the effective productivity of domestic low-skilled labor.

Interpretation: The paper's key novelty is the productivity effect. Without it, offshoring looks like a labor-supply shock; with it, offshoring can raise the wage of the very factor whose tasks are moved offshore.

2.4 Contribution to policy interpretation

- The model implies that opposition to offshoring based only on labor-supply displacement may miss an important source of gains.
- It also warns that the gains are not guaranteed. Relative-price and labor-supply effects can offset the productivity effect in some environments.
- The paper therefore provides a framework for thinking about when offshoring creates shared gains and when it creates distributional conflict.

Interpretation: The contribution is not the claim that all offshoring benefits all workers. The contribution is a decomposition that clarifies the conditions under which offshoring can benefit the factor whose tasks are offshored.

3 Model primitives and measurement

3.1 Goods, factors, and tasks

- The home country can produce two goods, X and Y , under constant returns to scale.
- Production of each good requires a continuum of tasks performed by low-skilled labor, a continuum of tasks performed by high-skilled labor, and possibly tasks performed by other factors.
- Tasks performed by low-skilled labor are called L -tasks; tasks performed by high-skilled labor are called H -tasks.
- The measure of tasks for each factor is normalized to one.

- Low-skill tasks are indexed by $i \in [0, 1]$ and ordered by increasing difficulty of offshoring.

Interpretation: The model separates who performs a task from where the task is performed. This is the central modeling move that makes task trade different from goods trade.

3.2 Task intensity and input coefficients

- If an L -task is performed at home in industry j , it uses a_{Lj} units of domestic low-skilled labor per unit of output.
- If an H -task is performed at home in industry j , it uses a_{Hj} units of domestic high-skilled labor per unit of output.
- The model can allow substitution between L -tasks and H -tasks, so a_{Lj} and a_{Hj} can be choice variables selected to minimize unit cost.
- The paper assumes good X is skill intensive, so $a_{Hx}/a_{Lx} > a_{Hy}/a_{Ly}$.

Interpretation: The relative task intensities determine which good is labor intensive and which is skill intensive, which matters for the relative-price effect.

3.3 Offshoring costs

- If a home firm performs low-skill task i abroad, it requires $a_{Lj}\beta t(i)$ units of foreign low-skilled labor.
- $t(i)$ is increasing in i , so tasks with lower indexes are easier to offshore.
- β is a technology shifter. A fall in β represents better communication, transportation, coordination, or monitoring technology.
- The benchmark case assumes $t_x(i) = t_y(i) = t(i)$, so the schedule of offshoring costs is the same across industries.

Interpretation: Heterogeneous task costs generate a marginal task. Improvements in offshoring technology move that margin and create cost savings on all inframarginal offshore tasks.

3.4 Marginal offshore task

Let w be the home low-skill wage and w^* the foreign low-skill wage. The marginal task I satisfies

$$w = \beta t(I)w^*. \quad (1)$$

Tasks with $i < I$ are offshored; tasks with $i > I$ are performed at home.

Interpretation: The marginal task condition says that firms offshore tasks until the wage savings are exactly offset by the task-specific offshoring cost.

3.5 Average cost of low-skill task performance

The paper defines

$$\Omega(I) = 1 - I + \frac{\int_0^I t(i)di}{t(I)}. \quad (2)$$

Because $t'(i) > 0$, $\Omega(I) < 1$ when $I > 0$.

Interpretation: $w\Omega(I)$ is the effective average cost of performing the full unit measure of low-skill tasks after optimal offshoring. When Ω falls, low-skilled labor becomes effectively more productive.

4 Core model and theoretical specifications

4.1 Zero-profit conditions

With optimal offshoring of low-skill tasks, the unit cost of good j can be written as

$$p_j \leq wa_{Lj}(\cdot)\Omega(I) + sa_{Hj}(\cdot) + \dots, \quad (3)$$

with equality if good j is produced.

Interpretation: This representation is the first key result. Offshoring changes production costs through $\Omega(I)$, so it is as if low-skilled labor had become more productive by a factor $1/\Omega(I)$.

4.2 Factor-market clearing

Domestic low-skilled labor clears according to

$$a_{Lx}(\cdot)x + a_{Ly}(\cdot)y = \frac{L}{1-I}. \quad (4)$$

Domestic high-skilled labor clears according to

$$a_{Hx}(\cdot)x + a_{Hy}(\cdot)y = H. \quad (5)$$

Interpretation: Offshoring also expands the effective supply of domestic low-skilled labor because home workers perform only the fraction $1 - I$ of low-skill tasks left at home.

4.3 The three wage effects

Differentiating the equilibrium system yields the low-skill wage decomposition

$$\hat{w} = -\hat{\Omega} + \mu_1\hat{p} - \mu_2\frac{dI}{1-I}. \quad (6)$$

- $-\hat{\Omega}$ is the **productivity effect**. Since a fall in offshoring costs reduces Ω , this term raises w .
- $\mu_1\hat{p}$ is the **relative-price effect**. If offshoring changes world relative supplies and terms of trade, wages respond through Stolper–Samuelson forces.
- $-\mu_2 dI/(1-I)$ is the **labor-supply effect**. Offshoring releases domestic low-skilled workers from tasks that move abroad.

The high-skill wage has an analogous decomposition:

$$\hat{s} = -\mu_3\hat{p} + \mu_4\frac{dI}{1-I}. \quad (7)$$

Interpretation: The decomposition is the central analytical tool of the paper. It shows exactly why offshoring can raise or lower the wage of the offshored factor depending on the environment.

5 The productivity effect

5.1 Small Heckscher–Ohlin economy

In a small open economy with two goods and two factors, relative goods prices and foreign wages are fixed. If both goods are produced, the zero-profit conditions are

$$1 = \Omega w a_{Lx}(\Omega w/s) + s a_{Hx}(\Omega w/s), \quad (8)$$

$$p = \Omega w a_{Ly}(\Omega w/s) + s a_{Hy}(\Omega w/s). \quad (9)$$

These equations determine Ωw and s independently of β .

Interpretation: Since Ωw is fixed, a fall in Ω must raise w . Offshoring low-skill tasks raises the real wage of low-skilled workers in this setting.

5.2 Why the productivity effect exists

- When β falls, firms save on all tasks that are already offshore.
- These inframarginal cost savings are proportional to the share of low-skill tasks in production costs.
- The savings raise productivity in the labor-intensive sector more than in the skill-intensive sector.
- In a small Heckscher–Ohlin economy with fixed goods prices, factor prices adjust so the low-skilled wage rises.

Interpretation: The effect is absent when $I = 0$ because there are no inframarginal offshore tasks. It becomes more important as offshoring volume grows.

5.3 Offshoring versus immigration

- Immigration increases the domestic supply of low-skilled labor.
- Offshoring reduces the cost of performing low-skill tasks and creates productivity gains for firms.
- Immigration therefore lacks the inframarginal productivity effect that task offshoring creates.

Interpretation: This is why offshoring cannot be analyzed simply as if it were an increase in foreign labor supply.

6 Relative-price and labor-supply effects

6.1 Large economy and relative goods prices

In a large economy, the home country affects world relative supply. The foreign economy has analogous zero-profit conditions:

$$1 = A^* w^* a_{Lx}(w^*/s^*) + A^* s^* a_{Hx}(w^*/s^*), \quad (10)$$

$$p = A^* w^* a_{Ly}(w^*/s^*) + A^* s^* a_{Hy}(w^*/s^*). \quad (11)$$

World outputs satisfy equations of the form

$$x + x^* = \frac{a_{Ly}(H + H^*/A^*) - a_{Hy}(L^*/A^* + L/\Omega)}{\Delta_a}, \quad (12)$$

$$y + y^* = \frac{a_{Hx}(L^*/A^* + L/\Omega) - a_{Lx}(H + H^*/A^*)}{\Delta_a}. \quad (13)$$

Interpretation: A fall in Ω raises the effective world supply of low-skilled labor used by home firms, increasing the relative supply of labor-intensive goods and potentially lowering their relative price.

6.2 Stolper–Samuelson channel

- If the labor-intensive good becomes more abundant, its relative price tends to fall.
- A fall in the relative price of the labor-intensive good reduces the low-skill wage relative to the high-skill wage in standard Heckscher–Ohlin logic.
- This relative-price effect can offset the productivity effect.

Interpretation: Offshoring can benefit low-skilled workers through productivity while hurting them through relative prices. The net result depends on which force is stronger.

6.3 Specialized economy and labor-supply effect

If the home country produces only good X , then

$$a_{Lx}(w\Omega/s)x = \frac{L}{1-I}, \quad (14)$$

$$a_{Hx}(w\Omega/s)x = H. \quad (15)$$

The low-skill wage satisfies

$$\hat{w} = -\hat{\Omega} - \frac{1 - \theta_{Lx}}{\sigma_x} \frac{dI}{1-I}. \quad (16)$$

Using the definitions of $\eta(I)$ and $\gamma(I)$, this becomes

$$\hat{w} = \left[\eta\gamma - \frac{1 - \theta_{Lx}}{\sigma_x} \right] \frac{dI}{1-I}. \quad (17)$$

Interpretation: The first term is the productivity effect. The second term is the labor-supply effect. The first bit of offshoring can reduce low-skill wages because there are no inframarginal offshore tasks yet, but further reductions in offshoring costs can raise wages once $I > 0$.

7 Offshoring skill-intensive tasks

7.1 Separate task margins for low-skill and high-skill tasks

Let I_L and I_H be the marginal offshored tasks for low-skilled and high-skilled labor. The two marginal conditions are

$$w = w^* \beta_L t_L(I_L), \quad (18)$$

$$s = s^* \beta_H t_H(I_H). \quad (19)$$

With both goods produced, zero-profit conditions become

$$1 = \Omega_L w a_{Lx}(\Omega_L w / \Omega_H s) + \Omega_H s a_{Hx}(\Omega_L w / \Omega_H s), \quad (20)$$

$$p = \Omega_L w a_{Ly}(\Omega_L w / \Omega_H s) + \Omega_H s a_{Hy}(\Omega_L w / \Omega_H s). \quad (21)$$

Interpretation: Each factor now has its own offshoring margin and its own productivity effect.

7.2 Factor-specific productivity effects

If the country remains incompletely specialized, then the zero-profit equations determine $\Omega_L w$ and $\Omega_H s$ independently of the offshoring cost shifters. Hence

$$\hat{w} = -\hat{\Omega}_L, \quad (22)$$

$$\hat{s} = -\hat{\Omega}_H. \quad (23)$$

Interpretation: Improvements in offshoring technology for high-skill tasks raise skilled wages through the same logic that improvements for low-skill tasks raise low-skilled wages.

7.3 Symmetric offshoring of low- and high-skill tasks

If $t_L(i) = t_H(i)$ and $\beta_L = \beta_H$, then low-skill and high-skill tasks are offshored in the same fraction. This is analogous to Hicks-neutral productivity growth in both industries when the home country is small.

Interpretation: In a large economy, the relative-price effect can still matter because the expansion in productivity changes relative world output supplies.

8 Theoretical design and limitations

8.1 What the paper proves cleanly

- A fall in task-offshoring costs can be represented as factor-augmenting technical change.
- The wage effect can be decomposed into productivity, relative-price, and labor-supply channels.
- The productivity effect is positive for the factor whose tasks are offshored whenever there is an existing positive mass of offshored tasks.
- The relative-price effect and labor-supply effect can offset or dominate the productivity effect.

8.2 What the paper abstracts from

- It abstracts from firm heterogeneity, search frictions, contracting problems, and organizational boundaries.
- It treats offshoring costs as exogenous and ordered by task difficulty.
- It uses competitive markets and constant returns.
- It does not estimate offshoring costs or test the decomposition empirically.

Interpretation: The model is deliberately simple. Its goal is not to capture every institutional detail of offshoring but to isolate the general-equilibrium forces that standard models miss.

9 Tables, figures, and original equation panels

9.1 Panel 1: Core offshoring condition and unit cost

Read:

- Equation (1) defines the marginal offshored task I .
- Equation (2) gives the raw unit-cost expression before substituting the marginal condition.

Economic message: Firms offshore tasks until wage savings exactly offset offshoring costs.

$$(1) \quad w = \beta t(I) w^*.$$

In a competitive economy, the price of any good is less than or equal to the unit cost of production, with equality whenever a positive quantity of the good is produced. The unit cost of producing good j is the sum of the wages paid to domestic low-skilled labor, the wages paid to foreign labor for tasks performed offshore, the wages paid to domestic skilled labor for the unit measure of H -tasks, and the payments to any other factors of production. Considering firms' optimal choices of intensity a_{Lj} , a_{Hj} , etc., and the optimal offshoring of L -tasks, we have

$$(2) \quad p_j \leq w a_{Lj}(\cdot)(1 - I) + w^* a_{Lj}(\cdot) \int_0^I \beta t(i) di + s a_{Hj}(\cdot) + \dots, \quad \text{for } j = x, y,$$

9.2 Panel 2: Effective cost and factor markets

Read:

- Equation (3) shows the simplified unit-cost expression using $\Omega(I)$.
- Equations (4) and (5) show factor-market clearing for low-skilled and high-skilled labor.
- The low-skill market-clearing condition contains $L/(1 - I)$, capturing the labor-supply effect.

Economic message: The same offshoring margin generates both the productivity effect through $\Omega(I)$ and the labor-supply effect through $1/(1 - I)$.

possibility that there are additional factors and additional tasks besides those performed by low-skilled and high-skilled labor.

By substituting for w^* using (1), we can rewrite (2) as

$$(3) \quad p_j \leq w a_{Lj}(\cdot) \Omega(I) + s a_{Hj}(\cdot) + \dots, \quad \text{for } j = x, y,$$

where

$$\Omega(I) \equiv 1 - I + \frac{\int_0^I t(i) di}{t(I)}.$$

The first term on the right-hand side of (3) is the total cost of the unit measure of L -tasks in light of the profit-maximizing geographic allocation of these tasks. Notice that this cost is proportional to the chosen (or technologically fixed) intensity of task performance, with proportionality factor $w\Omega(I)$. Thus, $w\Omega(I)$ is the average cost of the low-skilled labor used to perform L -tasks, while s is the average cost of the skilled labor used to perform H -tasks. These average factor costs are the arguments in the $a_{Hj}(\cdot)$ functions, because the tasks using a given factor are performed in fixed combination. Notice, too, that $t'(i) > 0$ for all $i \in [0, 1]$ implies that $\Omega(I) < 1$ for $I > 0$; i.e., offshoring reduces the wage bill in proportion to the cost of performing all L -tasks at home, as long as some tasks are performed abroad.

Next consider the domestic factor markets. The market for low-skilled labor clears when employment by the two industries in the tasks performed at home exhausts the domestic factor supply, L . Each firm completes a fraction $1 - I$ of L -tasks at home, and an L -task in industry j employs a_{Lj} units of labor per unit of output. Letting x and y denote the outputs of the two industries, we have $(1 - I)a_{Lx}(\cdot)x + (1 - I)a_{Ly}(\cdot)y = L$, or

$$(4) \quad a_{Lx}(\cdot)x + a_{Ly}(\cdot)y = \frac{L}{1 - I}.$$

This way of writing the market-clearing condition highlights the fact that offshoring leverages the domestic factor supply; i.e., that an expansion in I is like an increase in L . For skilled labor, H , we have the usual

$$(5) \quad a_{Hx}(\cdot)x + a_{Hy}(\cdot)y = H,$$

9.3 Panel 3: Wage-effect decomposition

Read:

- Equation (6) decomposes the low-skilled wage response into productivity, relative-price, and labor-supply effects.
- Equation (7) gives the analogous skilled-wage decomposition.

Economic message: This decomposition is the paper's main organizing framework.

in the wage of low-skilled labor as

$$(6) \quad \hat{w} = -\hat{\Omega} + \mu_1 \hat{p} - \mu_2 \frac{dI}{1-I},$$

where the μ_i 's collect all the terms that multiply dp/p and $dI/(1-I)$, respectively.

We call the first term on the right-hand side of (6) the *productivity effect*. As the technology for offshoring improves ($d\beta < 0$), the cost of performing the set of L -tasks declines in both industries ($\Omega < 0$).¹⁴ A firm's cost savings are proportional to its payments to low-skilled labor. These

Equation (6): low-skill wage decomposition

$$(7) \quad \hat{s} = -\mu_3 \hat{p} + \mu_4 \frac{dI}{1-I}.$$

Notice that there is no productivity effect. This is because a fall in β reduces firms' costs of non-

Equation (7): high-skill wage decomposition

9.4 Panel 4: Productivity effect in a small Heckscher–Ohlin economy

Read:

- Equations (8) and (9) show zero-profit conditions in a small economy.
- They pin down Ωw and s .
- If Ω falls, w rises.

Economic message: In this environment, offshoring low-skill tasks is exactly like low-skilled-labor-augmenting technological progress.

Assuming that both industries are active in equilibrium, the zero-profit conditions imply¹⁶

$$(8) \quad 1 = \Omega w a_{Lx}(\Omega w/s) + s a_{Hx}(\Omega w/s)$$

and

$$(9) \quad p = \Omega w a_{Ly}(\Omega w/s) + s a_{Hy}(\Omega w/s).$$

Here, we have made explicit the dependence of the production techniques on the relative average factor costs, $\Omega w/s$, in view of the profit-maximizing choice of offshoring dictated by (1). Since the industries differ in factor intensities, these two equations uniquely determine Ωw and s , independently of β . Thus, as β falls, $\hat{w} = -\hat{\Omega}$ and $\hat{s} = 0$. We conclude that the productivity effect is the *only* effect that operates in the present setting.¹⁷ The relative-price effects are absent ($\mu_1 = \mu_3 = 0$), because terms of trade are exogenous in a small economy. The labor-supply effects are absent ($\mu_2 = \mu_4 = 0$), because factor prices are insensitive to factor supplies (at given commodity prices) in an economy with equal numbers of primary factors and produced goods.

We can compute the magnitude of the productivity effect by combining $\hat{w} = -\hat{\Omega}(I)$ and $\hat{w} = \hat{\beta} + \hat{f}(I)$, which follows from (1) and the fact that w^* is fixed for a small country. Solving this pair of equations gives

$$\hat{w} = \int_0^I t(i) di$$

9.5 Panel 5: Relative-price effects in a large economy

Read:

- Equations (10) and (11) describe foreign zero-profit conditions.
- Equations (12) and (13) describe world output supplies.
- Changes in Ω affect relative world output and therefore relative goods prices.

Economic message: The relative-price effect reintroduces the standard Stolper–Samuelson logic into the task-trade framework.

firms in both industries. This means that, were a foreign firm to perform all tasks at the same intensities as a domestic firm, its output would be only $1/A^*$ times as great. Assuming incomplete specialization in the foreign country, the zero-profit conditions for indigenous foreign firms imply

$$(10) \quad 1 = A^* w^* a_{Lx}(w^*/s^*) + A^* s^* a_{Hx}(w^*/s^*)$$

and

$$(11) \quad p = A^* w^* a_{Ly}(w^*/s^*) + A^* s^* a_{Hy}(w^*/s^*).$$

Comparing (8) and (9) with (10) and (11), we see that incomplete specialization in both countries implies “adjusted factor price equalization”; that is, $w\Omega = w^*A^*$ and $s = s^*A^*$.

In such an equilibrium, home firms choose their production techniques based on the relative average factor costs $w\Omega/s$. Foreign firms choose theirs based on the relative factor prices w^*/s^* . Therefore, with adjusted factor price equalization, the relative cost-minimizing techniques are the same in the two countries, i.e., $a_{ij}A^* = a_{ij}^*$. The foreign factor-market clearing conditions can be written as

$$A^* a_{Lx}x^* + A^* a_{Ly}y^* + \beta \int_0^l r(i) di (a_{Lx}x^* + a_{Ly}y^*) = L^*$$

and

$$A^* a_{Hx}x^* + A^* a_{Hy}y^* = H^*,$$

Foreign zero-profit and factor-market conditions

$$(12) \quad x + x^* = \frac{a_{Ly} \left(H + \frac{H^*}{A^*} \right) - a_{Hy} \left(\frac{L^*}{A^*} + \frac{L}{\Omega} \right)}{\Delta_a}$$

and

$$(13) \quad y + y^* = \frac{a_{Hx} \left(\frac{L^*}{A^*} + \frac{L}{\Omega} \right) - a_{Lx} \left(H + \frac{H^*}{A^*} \right)}{\Delta_a},$$

where $\Delta_a = a_{Hy}a_{Lx} - a_{Lx}a_{Hy} > 0$.
Equilibrium in the goods market requires

World output equations

9.6 Panel 6: Labor-supply effect in a specialized economy

Read:

- Equations (14) and (15) are factor-market clearing conditions when the home country produces only good X .
- Equation (16) decomposes the low-skilled wage response into productivity and labor-supply terms.
- The η and γ definitions clarify when the productivity effect can dominate the labor-supply effect.

Economic message: Offshoring can initially reduce low-skill wages but later raise them if the productivity effect becomes sufficiently large.

tion for this industry implies that equation (8) must hold, whereas the price p is less than the unit cost of production in industry Y . The factor-market clearing conditions are quite simple in this setting, and they require

$$(14) \quad a_{Lx}(w\Omega/s)x = \frac{L}{1-I},$$

and

$$(15) \quad a_{Hx}(w\Omega/s)x = H.$$

Consider a decline in the cost of trading tasks ($d\beta < 0$). Differentiating (8) gives

$$\theta_{Lx}(\hat{w} + \hat{\Omega}) + (1 - \theta_{Lx})\hat{s} = 0,$$

while differentiating the ratio of (14) to (15) implies

$$\sigma_x(\hat{s} - \hat{w} - \hat{\Omega}) = \frac{dI}{1-I},$$

Specialized economy and equation (16)

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$$(16) \quad \hat{w} = -\hat{\Omega} - \frac{1 - \theta_{Lx}}{\sigma_x} \frac{dI}{1-I}.$$

The first term on the right-hand side of (16) is the productivity effect, as before. The second term is the labor-supply effect on low-skilled wages. The former effect is positive, while the latter is negative and reflects the adjustment in wages necessary for all domestic low-skilled workers to be employed when performing the smaller set of tasks undertaken in the home country.

To compare the magnitudes of these offsetting effects, we need to relate $-\hat{\Omega}$ to $dI/(1-I)$. This can easily be done using the definition of $\Omega(I)$ or the derivative $d\Omega/dI$ reported in footnote 14. We find that $-\hat{\Omega} = \eta\gamma dI/(1-I)$, and so

$$\hat{w} = \left[\eta\gamma - \frac{1 - \theta_{Lx}}{\sigma_x} \right] \frac{dI}{1-I}$$

where, as before,

$$\eta(I) \equiv \frac{r'(I)(1-I)}{r(I)}$$

is the elasticity of the trade cost schedule when expressed as a function of $1-I$, and

$$\gamma(I) \equiv \frac{\int_0^I r(i) di}{\int_0^I r(i) di + (1-I)r(I)}$$

Elasticities η and γ and wage condition

9.7 Panel 7: Offshoring high-skill tasks

Read:

- Equations (17) and (18) define marginal tasks for low- and high-skilled labor.
- Equations (19) and (20) give the zero-profit conditions when both factors' tasks can be offshored.
- The resulting wage changes are $\hat{w} = -\hat{\Omega}_L$ and $\hat{s} = -\hat{\Omega}_H$ in the small open economy.

Economic message: The productivity-effect logic applies to any factor whose tasks can be traded.

Now, I_L is the marginal task using factor f that is performed offshore. For low-skilled labor, we have

$$(17) \quad w = w^* \beta_L I_L(I_L),$$

as before. The analogous condition for high-skilled labor is

$$(18) \quad s = s^* \beta_H I_H(I_H).$$

If home firms produce both goods, the zero-profit conditions imply

$$(19) \quad 1 = \Omega_L w a_{Lx}(\Omega_L w / \Omega_H s) + \Omega_H s a_{Hx}(\Omega_L w / \Omega_H s)$$

and

$$(20) \quad p = \Omega_L w a_{Ly}(\Omega_L w / \Omega_H s) + \Omega_H s a_{Hy}(\Omega_L w / \Omega_H s),$$

where

Equations (17)–(20): offshoring low- and high-skill tasks

Now, in fact, (17) and (18) determine w and s independently of p_L and p_H . Therefore, as long as the country remains incompletely specialized, a fall in the cost of offshoring one or both types of task leaves $\Omega_L w$ and $\Omega_H s$ unchanged. It follows that

$$\hat{w} = -\hat{\Omega}_L$$

and

$$\hat{s} = -\hat{\Omega}_H,$$

with $d\Omega_L/dR_L > 0$, $d\Omega_H/dR_H = 0$, $d\Omega_L/dR_H > 0$ and $d\Omega_H/dR_L = 0$. That is, an improvement

Factor-specific wage effects

10 Short summary

- The paper develops a theory of offshoring as trade in tasks.
- Each factor performs a continuum of tasks, and tasks differ in offshoring costs.
- Firms offshore low-cost tasks until the marginal task satisfies $w = \beta(I)w^*$.
- Offshoring lowers the effective cost of low-skill tasks through $\Omega(I)$.
- The low-skill wage effect decomposes into a productivity effect, a relative-price effect, and a labor-supply effect.
- In a small two-good, two-factor economy, only the productivity effect operates, so low-skilled wages rise when low-skill offshoring costs fall.
- In a large economy, relative-price changes can offset the productivity effect.
- In a specialized economy, the labor-supply effect can offset the productivity effect, especially at the start of offshoring.
- If high-skill tasks can be offshored, skilled workers receive an analogous productivity effect.
- The main contribution is to show that offshoring is not necessarily harmful to the factor whose tasks move abroad; it can raise that factor's wage by augmenting productivity.

11 Conclusion

11.1 Core conclusion

Grossman and Rossi-Hansberg show that trade in tasks changes the logic of globalization. A fall in offshoring costs is not equivalent to a fall in goods trade costs or to an increase in foreign labor supply. It creates a productivity effect that can benefit the domestic factor whose tasks are offshored.

11.2 Economic intuition

When some tasks are already performed abroad, a further decline in offshoring costs reduces the cost of all inframarginal offshore tasks. Domestic firms become more productive in using the corresponding domestic factor. In a small open economy, this raises the wage of that factor. In richer environments, relative-price and labor-supply effects may offset the productivity gain.

11.3 Takeaway

The paper provides the canonical task-trade framework for thinking about offshoring. Its lasting insight is that the distributional effects of globalization depend not only on what goods are traded, but on which tasks are traded and how task trade changes effective factor productivity.